

Dear Editor,

Thank you very much for arranging to review our manuscript. However, it is a bit surprising that despite answering all the questions/queries of Referee 1, a disjoint/new set of queries has been raised, which no mention and no relation whatsoever to the earlier set. **How exactly has my earlier response been accounted for and why am I supposed to address this new set without a resolution of my earlier response and revisions?** I believe the journal would highly benefit from improving such editorial service.

No reference to Referee 1's questions and to my answers show that Referee 2 has been unaware of the earlier round. Hence I attach my response to the earlier round along with this one.

Despite the highly unusual treatment of my revisions of the manuscript, I am addressing this new set of questions/suggestions raised/made by a different referee, namely Referee 2. Detailed responses to the comments of the referee have been provided below (Referee's comments are highlighted in blue). In view of these changes, I trust the manuscript will now be handled in a proper way and judged suitable for publication without further long delays.

Sincerely,
Author

Response to questions/comments of the Referee 2:

1. Although candidate materials (e.g., RhSi, CoSi) are mentioned, it would significantly strengthen the manuscript if the author could provide quantitative estimates for experimentally relevant parameter regimes. In particular, typical values of chemical potential, magnetic field strength, scattering rates, and the expected magnitude of the predicted effects in real materials would help assess experimental feasibility.

In the manuscript, I had already provided reference for realistic materials, which thus give us experimentally-feasible regimes:

"The parameter values have been taken from Ref. [1]. There, $\hbar v_F = 1.23 \text{ eV \AA}$, which translates into

$$v_F = \frac{1.23 \times 10^{-10} \text{ m eV}}{6.58 \times 10^{-16} \text{ eV s}} = 0.1869 \times 10^6 \text{ m/s} \simeq 0.06 \times 10^{-2} c,$$

where c is the speed of light. Therefore, in natural units, we set the value of v_F to 6×10^{-4} ." The referee seems to have missed it.

The chemical potential used here is the one measured with respect to the band-crossing point, taken to be the zero of the chemical potential (well-used notation in the literature) within the range 0.1–0.2 eV. Varying μ beyond such ranges is unrealistic when compared to actual material parameters. In fact, going beyond such ranges, two things will happen: (1) The low-energy excitations in the vicinity of the RSWN will no longer be applicable. Higher energy windows (i.e., larger magnitudes of μ) will involve band-bending, when the continuum model very near the RSWN at the Γ -point will no longer be a reasonable modelling of the system. (2) RSWN appears in chiral crystals with strong spin-orbit couplings, which host multiple distinct types of band crossings, albeit at different energy levels. When we increase the magnitude of μ considerably, contributions from other kinds of nodes will become dominant, when the Fermi level will tune to their nodal points. In some intermediate ranges, there might be significant internode scatterings as well. This leads to a very complex scenario which is beyond the scope of this paper, for which continuum models may not even be justified, necessitating the use of actual lattice models. We refer the referee to our discussions in the introduction, which sheds light on these facts:

"For example, in RhSi, there exist fourfold- and sixfold-degenerate nodes at the Γ - and R -points of the BZ, carrying Chern numbers of ± 4 , and featuring the largest possible momentum-separation between the nodal-points' locations [2, 3]. While the former represents an RSWN, the bands at the latter are doubly-degenerate, featuring a pair of pseudospin-1 triple-point fermions (TPFs). The energy-offset between these nodes is $\sim 0.4 \text{ eV}$."

In contemporary experiments, scientists have reported using magnetic fields upto $\sim 10 \text{ T}$ – for example, see Ref. [4] [which was already cited in the manuscript]. Since 1 T (in SI units) is equivalent to $\sim 1950 \text{ eV}^2$, my plots adhere to this experimental regime. One could have plotted the curves in the manuscript for higher values of B , but is it not very useful as the nature of the curves remains the same. I have added this line:

"In contemporary experiments, scientists have reported using magnetic fields upto $\sim 10 \text{ T}$ [4], which is equivalent to $\sim 1950 \text{ eV}^2$ in natural units. Our representative plots depict the global behaviour of conductivity within this experimentally-feasible range."

Scattering rate is not an input parameter in my calculations going beyond the relaxation time — it is a self-consistent part appearing from solving the integro-differential equation, viz. Eq. (27).

The curves themselves depict the expected magnitude of the conductivity to be measured in experiments and agree in magnitude when the data for chiral crystals is considered [4].

2. The neglect of Landau quantization is justified in the weak-field regime. However, the manuscript would benefit from a brief estimate of the range of magnetic fields over which this approximation remains valid, especially in relation to realistic material parameters.

I have added the following discussions: “Let us provide an example regarding how to estimate the bounds on the allowed strength of $\mathbf{B}$ [5]. Let us consider a Weyl node subjected to a magnetic field oriented along the z -direction. Solving the the energy dispersion relation characterising the n_L -th Landau level (LL), we pick a level n_L carrying positive energy (i.e., $n_L > 0$), which takes the form of $\epsilon_{n_L}^s(k_z) = |s| v_F \sqrt{2eBn_L + (v_F k_z)^2}$. For the semiclassical treatment to remain self-consistent, the number of occupied LLs, N_L , must be sufficiently large (i.e., $N_L \gg 1$). Put differently, the energy gap, $\Delta\epsilon_s(B) \equiv |s| v_F \left(\sqrt{2eB(N_L+1)} - \sqrt{2eBN_L} \right)$, separating the topmost filled Landau level from the lowest empty one at $k_z = 0$, must remain negligibly small in comparison with the chemical potential: $\Delta\epsilon_s(B) \ll \mu$. A straightforward estimate of this energy gap proceeds as follows:

$$\Delta\epsilon_s(B) = |s| v_F \sqrt{2eBN_L} \left(\sqrt{1 + \frac{1}{N_L}} - 1 \right) \simeq v_F \sqrt{\frac{eB}{2N_L}}. \quad (1)$$

Noting that N_L denotes the index of the highest occupied LL, the chemical potential (μ) must satisfy $v_F \sqrt{2eBN_L} \leq \mu < |s| v_F \sqrt{2eB(N_L+1)}$ and $N_L = \left\lfloor \frac{1}{2eB} \left(\frac{\mu}{|s| v_F} \right)^2 \right\rfloor$, where $\lfloor x \rfloor$ denotes the floor function (i.e., the largest integer not exceeding x). Combining everything then yields the criterion,

$$B \ll \frac{1}{e} \left(\frac{\mu}{|s| v_F} \right)^2, \quad (2)$$

which must be satisfied for the applied magnetic field to be regarded as weak and for the validity of the semiclassical approximation to be ensured. We have used $\mu = 0.1$ and $v_F = 6 \times 10^{-6}$, which lead to an upper limit of the order of 10000 eV², which is well above ranges of B used in our plots.”

3. It appears that a phase-space factor (the $D_s(k)$ factor) may be missing in Eq. (30). The author should verify this expression and clarify whether the omission is intentional or a typographical oversight.

I thank the referee for pointing out this typo. I have now corrected it.

4. The study is based on a low-energy model of a single isolated node. In realistic materials, multiple nodes are typically present, and internode scattering can become relevant. The author should comment on how the inclusion of internode scattering might modify the longitudinal magnetoconductivity and whether the main conclusions are expected to remain robust.

Around the vicinity of each node, a different low-energy model is valid, depicting the nodal-point structure of that node. The chemical potential used here is the one measured with respect to the band-crossing point, taken to be the zero of the chemical potential (well-used notation in the literature) within the range 0.1–0.2 eV. Varying μ beyond such ranges is unrealistic when compared to actual material parameters. In fact, going beyond such ranges, two things will happen: (1) The low-energy excitations in the vicinity of the RSWN will no longer be applicable. Higher energy windows (i.e., larger magnitudes of μ) will involve band-bending, when the continuum model very near the RSWN at the Γ -point will no longer be a reasonable modelling of the system. (2) RSWN appears in chiral crystals with strong spin-orbit couplings, which host multiple distinct types of band crossings, albeit at different energy levels. When we increase the magnitude of μ considerably, contributions from other kinds of nodes will become dominant, when the Fermi level will tune to their nodal points. In some intermediate ranges, there might be significant internode scatterings as well. This leads to a very complex scenario which is beyond the scope of this paper, for which continuum models may not even be justified, necessitating the use of actual lattice models. We refer the referee to our discussions in the introduction, which sheds light on these facts:

“For example, in RhSi, there exist fourfold- and sixfold-degenerate nodes at the Γ - and R -points of the BZ, carrying Chern numbers of ± 4 , and featuring the largest possible momentum-separation between the nodal-points’ locations [2, 3]. While the former represents an RSWN, the bands at the latter are doubly-degenerate, featuring a pair of pseudospin-1 triple-point fermions (TPFs). The energy-offset between these nodes is ~ 0.4 eV.”

I have now added some more explanations as follows: “In crystal lattices, it is essential to satisfy the Nielsen-Ninomiya theorem [6], which is physically reflected by the fact that the Weyl nodes (with opposite chiralities) must always exist in pairs in the entire BZ. For the spinless WSMs, arising in achiral crystals (e.g., TaAs family [7]), such conjugate partners are typically (almost) degenerate in energy, due to the presence of mirror or other roto-inversion symmetries. Therein, charge-pumping is an important internode phenomenon, in the presence of external electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields, having nonvanishing collinear components. This is an embodiment of the chiral anomaly in the arena of condensed matter physics [8–11], where the analog of the well-known concept of spacetime chirality (of relativistic fermions) is realised in the momentum space. On the contrary, the oppositely-charged chiral

nodes in chiral crystals need not be degenerate in energy, because the conjugate nodes are not related by crystal symmetries. In fact, they are observed to have discernible separations in energy and momenta, with an isolated RSWN located at an intrinsic chemical potential. As such, internode-scattering-induced charge-pumping becomes unimportant, while enhancing other phenomena like quantised circular photogalvanic effect [12–18] and circular dichroism [19, 20]. Thus, a single node will be relevant here, when we tune the external chemical potential to lie near the intrinsic energy of the concerned nodal point, as the chirally-conjugate node is well-separated in energy and momentum (see, for example, Refs. [21, 22]). In other words, the energy-separation between the $\chi = \pm$ members of the pair is much larger than the temperature-scale and the chemical-potential variation of the experiments. Hence, as long as we stick to a variation of μ within ~ 0.2 eV, our results are robust, since internode scatterings will be inconsequential.”

5. While the numerical results are clearly presented, the physical mechanism underlying the observed behavior deserves further clarification. In particular, the author should provide a more detailed explanation of why interband scattering leads to a change in the curvature or even the sign of the magnetoconductivity.

This explanation was already provided in the manuscript. I repeat it here for the convenience of the referee:

“... (I) for a single band with $\beta_{\text{inter}} = 0$, the charge conservation must be satisfied individually for that particular band; (II) a nonzero interband-coupling demands that the net charge, from summing over the two bands, must be conserved. The observations serve to reinforce the recurrent phenomenon that the response from a single isolated band is fundamentally different from the case when the quasiparticles are free to scatter with those populating other bands. In fact, the same thing has been observed for the two bands of an isolated Kramers-Weyl node [23], where downward-curved parabolas morph into upward-curved ones for each of the bands, as soon as β_{inter} is turned on.”

I. LIST OF CHANGES

I have marked all additions in the manuscript with the colour red.

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- [1] B. Xu, Z. Fang, M.-Á. Sánchez-Martínez, J. W. F. Venderbos, Z. Ni, T. Qiu, K. Manna, K. Wang, J. Paglione, C. Bernhard, C. Felser, E. J. Mele, A. G. Grushin, A. M. Rappe, and L. Wu, Optical signatures of multifold fermions in the chiral topological semimetal CoSi, *Proceedings of the National Academy of Science* **117**, 27104 (2020).
 - [2] D. S. Sanchez, I. Belopolski, T. A. Cochran, X. Xu, J.-X. Yin, G. Chang, W. Xie, K. Manna, V. Süß, C.-Y. Huang, N. Alidoust, D. Multer, S. S. Zhang, N. Shumiya, X. Wang, G.-Q. Wang, T.-R. Chang, C. Felser, S.-Y. Xu, S. Jia, H. Lin, and M. Zahid Hasan, Topological chiral crystals with heliocoid-arc quantum states, *Nature* **567**, 500–505 (2019).
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 - [4] F. Balduini, A. Molinari, L. Rocchino, V. Hasse, C. Felser, M. Sousa, C. Zota, H. Schmid, A. G. Grushin, and B. Gotsmann, Intrinsic negative magnetoresistance from the chiral anomaly of multifold fermions, *Nature Communications* **15**, 6526 (2024).
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 - [11] I. Mandal, Chiral anomaly and internode scatterings in multifold semimetals, *Phys. Rev. B* **111**, 165116 (2025).
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Dear Editor,

Thank you very much for arranging to review our manuscript. We have addressed all the questions/suggestions raised/made by the referee. Detailed responses to the comments of the referee have been provided below (Referee's comments are highlighted in blue). In view of these changes, we trust the manuscript will now be judged suitable for publication.

Sincerely,
Author

Response to questions/comments of the Referee:

1. Since the central advance of the present work is the treatment of scattering beyond constant τ , it would significantly strengthen the manuscript to include a direct comparison between the constant- τ (RTA) result and the exact solution for the same model parameters. Such a comparison would make transparent: What changes qualitatively, What changes only quantitatively, Whether interband scattering produces a genuinely new transport signature rather than a renormalization of the RTA result.

To address this point succinctly, we have added an entire new section, entitled ‘‘Comparison with the results obtained from relaxation-time approximation’’, containing the following discussions:

‘‘In our earlier work [1], we obtained the results for planar-Hall conductivity via the naive application of the RTA. For the sake of comparison, we elaborate on the explicit expressions of the longitudinal magnetoconductivity reported therein. For a chemical potential $\mu > 0$ and in the $T \rightarrow 0$ limit, the final result takes the following form:

$$\bar{\sigma}_{zz}^s = \sigma_{zz}^{s,\text{dr}} + \sigma_{zz}^{s,\text{bc}} + \sigma_{zz}^{s,m},$$

$$\sigma_{zz}^{s,\text{dr}} = \frac{e^2 \tau \mu^2}{6 \pi^2 s v_0}, \quad \sigma_{zz}^{s,\text{bc}} = \frac{4 e^4 \tau s^5 v_0^3 B^2}{15 \pi^2 \mu^2}, \quad \sigma_{zz}^{s,m} = \frac{e^4 \tau s v_0^3 \mathcal{G}_s B^2}{30 \pi^2} \frac{5 \mathcal{G}_s - 9 s^2}{\mu^2}. \quad (1)$$

Here, τ is the momentum-independent relaxation time within the RTA framework. Adding up the three parts, the total gives us

$$\bar{\sigma}_{zz}^s = \frac{e^2 \tau \mu^2}{6 \pi^2 s v_0} + \frac{e^4 \tau s v_0^3 B^2}{30 \pi^2} \frac{8 s^4 - 9 s^2 \mathcal{G}_s + 5 \mathcal{G}_s^2}{\mu^2}. \quad (2)$$

We note that the two conduction bands (viz. for $s = 1/2$ and $s = 3/2$) at the node at the same chemical potential and, so, there is no scope of having interband scatterings, unlike when we employ beyond-the-RTA (aka exact) method. So the net nodal contribution just comes from the sum of $\bar{\sigma}_{zz}^{s=1/2}$ and $\bar{\sigma}_{zz}^{s=3/2}$. We find that $\sigma_{zz}^{s,\text{bc}}$ is positive for both bands. On the other hand, $\sigma_{zz}^{s,m}$ is positive for $s = 1/2$ and negative for $s = 3/2$.

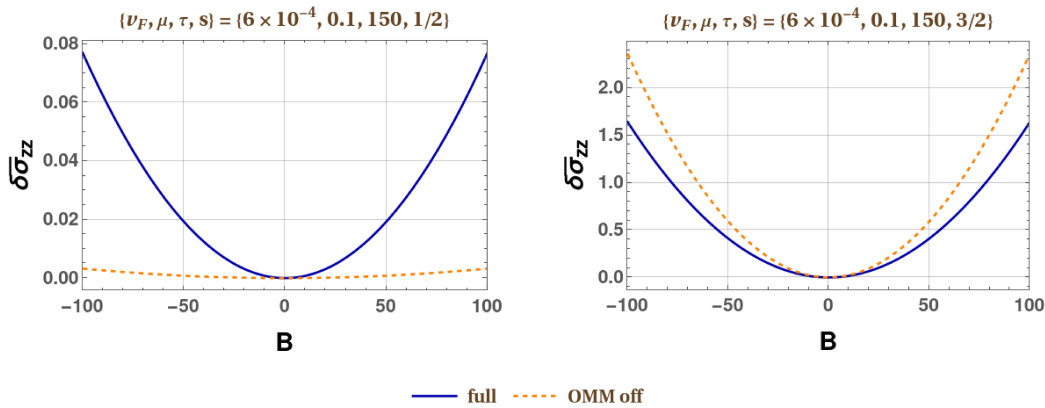


FIG. 4. $\delta\bar{\sigma}_{zz}(s)$ from each of the bands, considering the RTA expression shown in Eq. (1). While the dark-blue curves depict the variation of the full conductivity with respect to B (in eV^2) when OMM is taken into account, the orange curves represent the conductivity-versus- B profiles when OMM is switched off. Here, μ is in eV and τ is in eV^{-1} .

Let us define

$$\delta\bar{\sigma}_{zz}(s) \equiv \sigma_{zz}^s(B)/\sigma_{zz}^s(B=0) - 1. \quad (3)$$

For Fig. 4 shows the $\delta\bar{\sigma}_{zz}$ -curves for the two bands, using the same parameter values as Fig. 2 and setting $\tau = 150 \text{ eV}^{-1}$. Comparing the RTA curves with the exact-calculation curves (viz. Figs. 4 and 2), we infer that the naive RTA

approach fails to capture the actual characteristics that have been obtained by the exact solutions of the Boltzmann equations. We summarise our observations below: Comparing the RTA curves with the exact-calculation curves (viz. Figs. 4 and 2), we infer that the naive RTA approach fails to capture the actual characteristics that have been obtained by the exact solutions of the Boltzmann equations. We summarise our observations below:

- (a) For $s = 1/2$, the total conductivity has the same sign but values differing by orders of magnitude. The more dramatic effect is seen when OMM is not included — the curves are flipped. In the RTA case, the BC-only and OMM-induced parts have the same sign, with the magnitude of the latter dominating over the former. Whereas we have found that, in the exact-calculation case, the BC-only and OMM-induced parts have opposite signs with the magnitude of the former dominating over the latter.
- (b) For $s = 3/2$, the total conductivity itself is seen to have opposite signs for the two methods. For the exact-solution case, the BC-only and the OMM-induced parts are both negative and reinforce each other. As discussed earlier, for the RTA results, the BC-only and the OMM-induced parts are positive and negative, respectively, with the magnitude of the former dominating over the latter. So, in this case, the RTA is seen to miserably fail even to produce any trustworthy qualitative features.
- (c) We find that the interband scatterings, which are possible to incorporate only with the exact-solution method, change the single-band results drastically, introducing entirely new features starting at the qualitative level. Saliiently, depending on the value of the ratio, $\beta_{\text{inter}}/\beta_{\text{intra}}^{s,s}$, the parabolic conductivity-curves for $s = 3/2$ can be oriented upward or downward [cf. Fig. 3(a)]. Of course, such drastic variations cannot be observed from the RTA calculations, due to the absence of the possibility of interband scatterings.

”

2. From an experimental standpoint, an important question also remains open: can longitudinal magnetoconductivity distinguish an RSW node from a conventional Weyl node, or from generic multiband metallic behavior?

Since an RSW node consists of multiple linear bands meeting at a single point, it would be valuable to clarify whether: The predicted contains a robust and identifiable signature of the multifold character, Or whether the response could be interpreted as a superposition of conventional Weyl contributions.

From the added discussions for query 1, clearly the response is distinct from a mere superposition of conventional Weyl contributions. In particular, the multifold character introduces the possibility of interband scatterings (not captured within the RTA framework), which is manifested by the conductivity not being equal to the sum of the curves with β_{inter} set to zero. Crucially, depending on the dominant scattering mechanism, we can tune the value of $\beta_{\text{inter}}/\beta_{\text{intra}}^{s,s}$, which can lead to differing shapes of the the conductivity curves, as observed in Fig. 3(a). Such a possibility does not arise for the case of Weyl semimetals, which were studied in Ref. [2].

As for the question of distinctive response from different types of multifold nodes, the eigenspinors and the topological quantities will generically be different, which will give rise to different characteristics of the conductivity curves, both qualitatively and quantitatively. In particular, the difference in eigenspinors give rise to distinct forms of the quantity $\mathcal{M}_{s,\tilde{s}}$ appearing in the collision integrals, leading to distinct functional forms of the overlap functions [viz. Eq. (23)] and the relaxation times as well [viz. Eq. (25)]. Two such distinct examples can be readily found in Refs. [3, 4].

We have added the above discussions in the manuscript.

3. A brief discussion of experimentally accessible distinguishing features — for example scaling with chemical potential would considerably enhance the impact of the work.

Usually, the experiments involve measuring the magnetoresistance as functions of magnetic field and temperature, as done in Ref. [5] for a chiral crystal hosting pseudospin-1 nodes (with a threefold degeneracy). Hence, we have shown the variation with B in our plots.

Nevertheless, we tried varying μ within the range 0.1—0.2 eV, and we did not find any discernible variation in the conductivity and, practically, we got straight lines. Varying μ beyond such ranges is unrealistic when compared to actual material parameters. In fact, going beyond such ranges, two things will happen: (1) The low-energy excitations in the vicinity of the RSWN will no longer be applicable. Higher energy windows (i.e., larger magnitudes of μ) will involve band-bending, when the continuum model very near the RSWN at the Γ -point will no longer be a reasonable modelling of the system. (2) RSWN appears in chiral crystals with strong spin-orbit couplings, which host multiple distinct types of band crossings, albeit at different energy levels. When we increase the magnitude of μ considerably, contributions from other kinds of nodes will become dominant, when the Fermi level will tune to their nodal points. In some intermediate ranges, there might be significant internode scatterings as well. This leads to a very complex scenario which is beyond the scope of this paper, for which continuum models may not even be justified, necessitating the use of actual lattice models. We refer the referee to our discussions in the introduction, which sheds light on these facts:

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- [1] R. Ghosh, F. Haidar, and I. Mandal, Linear response in planar Hall and thermal Hall setups for Rarita-Schwinger-Weyl semimetals, [Phys. Rev. B **110**, 245113 \(2024\)](#).
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